Performance Estimation Problems (PEPs)

Systematic, principled, and computer-aided approaches to the analysis and design of first-order optimization algorithms

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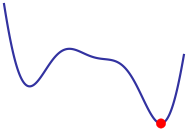
SMAI-MODE tutorial – March 2026

Context: numerical (continuous) optimization

Minimize $f : \mathbb{R}^d \to \mathbb{R}$ (e.g., with f continuous)

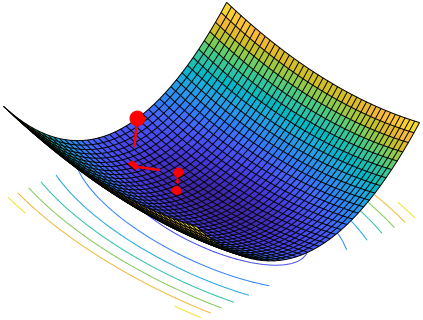
$$f(x_\star) \triangleq \min_{x \in \mathbb{R}^d} f(x).$$



Ubiquitous in applied mathematics and computer science.

Numerous applications for modeling (physics, economics), estimation (statistics, machine learning), decisions (control, operations research).

Usually solved via **iterative algorithm** generating sequence $x_0, x_1, \ldots, x_N$.



Gradient descent (stepsize α)

for $k = 0, 1, \ldots$ **do**

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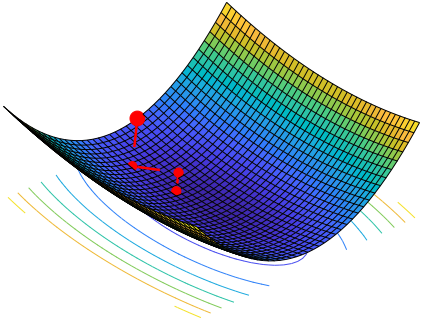

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For instance: **bounds** on certain notions of "error": $f(x_k) - f(x_\star)$, $\|x_k - x_\star\|$, $\|\nabla f(x_k)\|$, etc.

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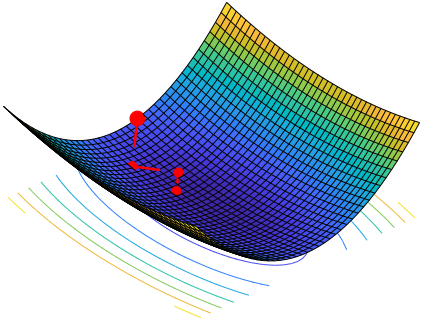

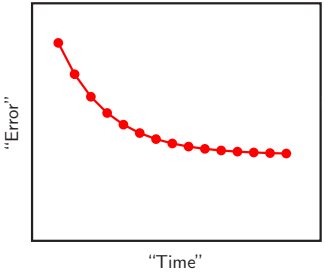

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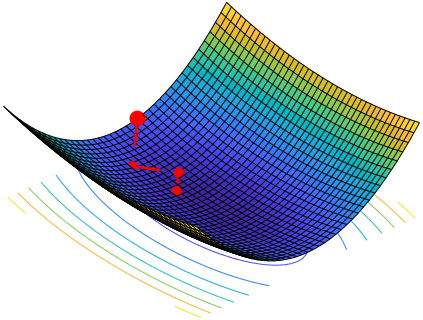

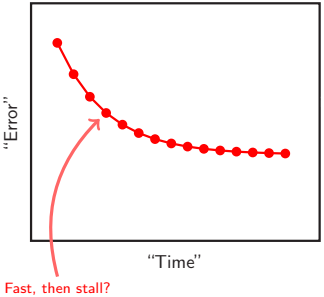

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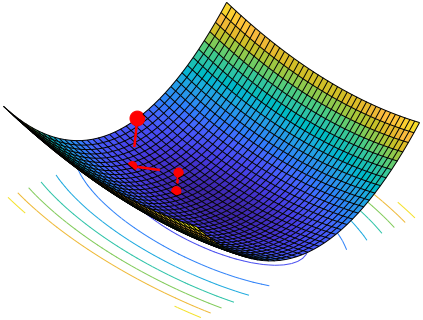

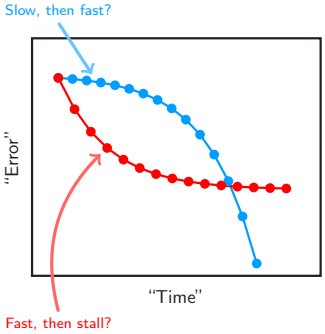

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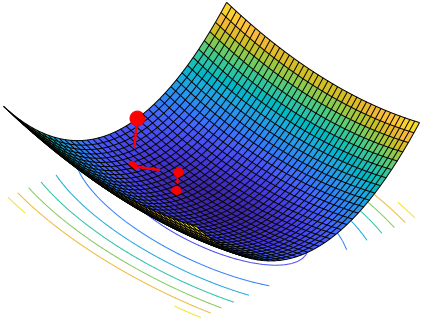

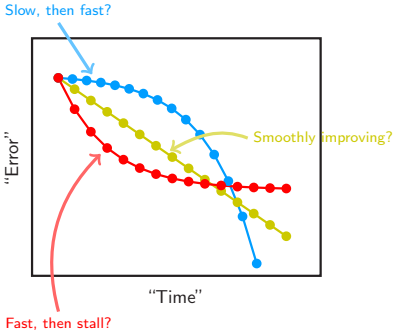

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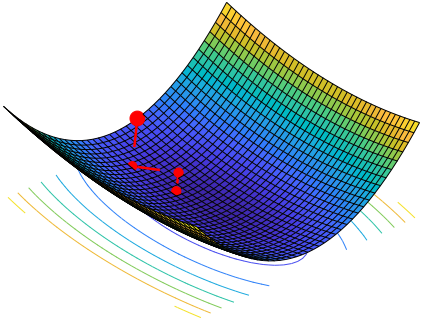

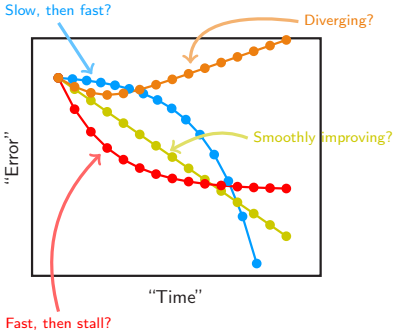

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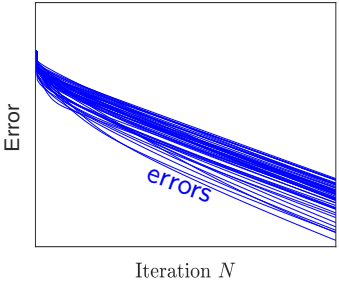

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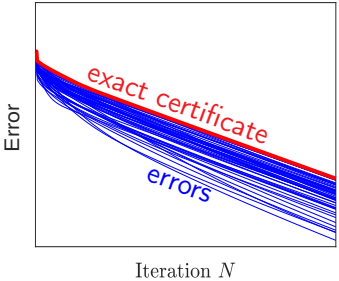

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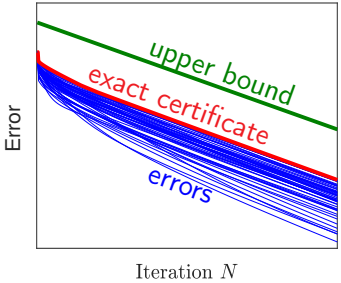

Example: analysis of a gradient method

Let $f : \mathbb{R}^d \to \mathbb{R}$ (continuously differentiable). Find $x_\star \in \mathbb{R}^d$ such that

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Question: what *a priori* guarantees after N iterations?

Examples: what about $f(x_N) - f(x_\star)$, $\|\nabla f(x_N)\|$, $\|x_N - x_\star\|$?

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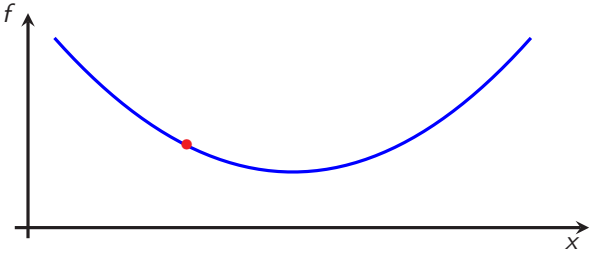

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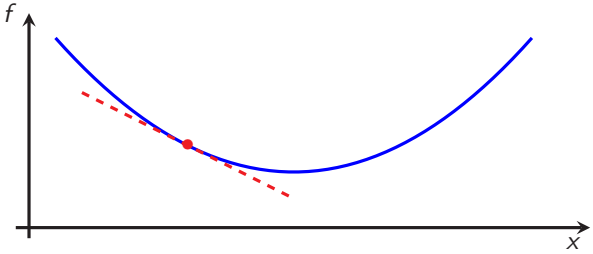

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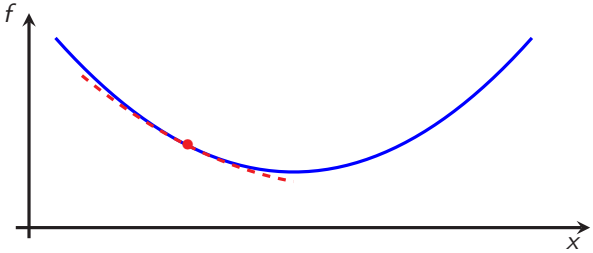

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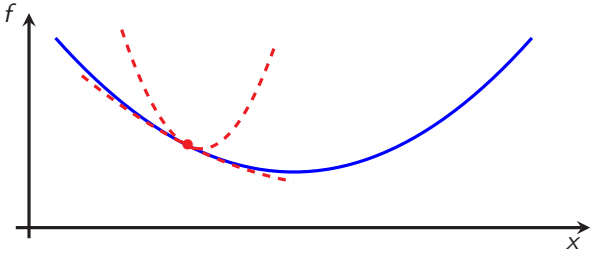

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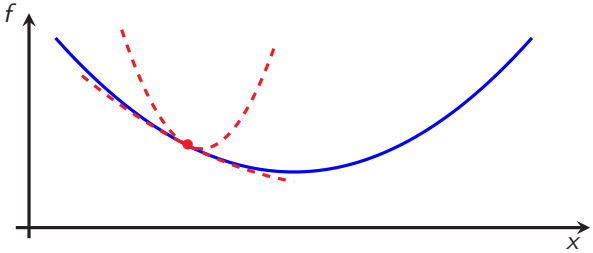

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(1&2) $\langle \nabla f(x) - \nabla f(y); x - y \rangle \geqslant \frac{1}{L+\mu}\|\nabla f(x) - \nabla f(y)\|^2 + \frac{\mu L}{L+\mu}\|x - y\|^2$.

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- Unique way to arrive to the desired result?
- How likely are we to find such proofs in more complicated cases?

Legitimate questions about performance analyses?

Lemma 3. *Assume that the function is L-smooth and μ strongly-convex and satisfies the strong-growth condition in Equation 10. Then, using the updates in Equation 3-5 and setting the parameters according to Equations 17-18 if $\eta \leq \frac{1}{\mu L}$, then the following relation holds:*

$$\mathbb{E}_{k+1}\gamma_k^2 \left[\mathbb{E}\|\nabla f(w_{k+1}) - f^*\right] \leq \frac{\alpha_k^2}{\rho\eta}\left(f(x_0) - f^*\right) + \frac{\eta_k^2}{2\rho\eta}\|x_0 - w^*\|^2 + \frac{\sigma^2\eta}{\rho}\sum_{t=0}^{k}\left[\gamma_t^2\eta_{t+1}^2\right]$$

Proof.

Let $r_{k+1} = \|w_{k+1} - w^*\|$, then using equation 5

$$r_{k+1}^2 = \|\beta_k v_k + (1-\beta_k)\zeta_k - w^* - \gamma_k\eta\nabla f(\zeta_k, z_k)\|^2$$
$$r_{k+1}^2 = \|\beta_k v_k + (1-\beta_k)\zeta_k - w^*\|^2 + \gamma_k^2\eta^2\|\nabla f(\zeta_k, z_k)\|^2 + 2\gamma_k\eta\langle w^* - \beta_k v_k - (1-\beta_k)\zeta_k, \nabla f(\zeta_k, z_k)\rangle$$

Taking expectation wrt to z_k,

$$\begin{aligned}
\mathbb{E}[r_{k+1}^2] &= \mathbb{E}\|\beta_k v_k + (1-\beta_k)\zeta_k - w^*\|^2 + \gamma_k^2\eta^2\mathbb{E}\|\nabla f(\zeta_k, z_k)\|^2 + 2\gamma_k\eta\mathbb{E}[\langle w^* - \beta_k v_k - (1-\beta_k)\zeta_k, \nabla f(\zeta_k, z_k)\rangle] \\
&\leq \|\beta_k v_k + (1-\beta_k)\zeta_k - w^*\|^2 + \gamma_k^2\eta^2\rho\|\nabla f(\zeta_k)\|^2 + 2\gamma_k\eta\langle w^* - \beta_k v_k - (1-\beta_k)\zeta_k, \nabla f(\zeta_k)\rangle + \gamma_k^2\eta^2\sigma^2 \\
&= \|\beta_k(v_k - w^*) + (1-\beta_k)(\zeta_k - w^*)\|^2 + \gamma_k^2\eta^2\rho\|\nabla f(\zeta_k)\|^2 + 2\gamma_k\eta\langle(w^* - \beta_k v_k - (1-\beta_k)\zeta_k, \nabla f(\zeta_k)\rangle + \gamma_k^2\eta^2\sigma^2 \\
&\leq \beta_k\|v_k - w^*\|^2 + (1-\beta_k)\|\zeta_k - w^*\|^2 + \gamma_k^2\eta^2\rho\|\nabla f(\zeta_k)\|^2 + 2\gamma_k\eta\langle(w^* - \beta_k v_k - (1-\beta_k)\zeta_k, \nabla f(\zeta_k)\rangle + \gamma_k^2\eta^2\sigma^2 \\
&\qquad \text{(By convexity of } \|\cdot\|^2\text{)} \\
&= \beta_k r_k^2 + (1-\beta_k)\|\zeta_k - w^*\|^2 + \gamma_k^2\eta^2\rho\|\nabla f(\zeta_k)\|^2 + 2\gamma_k\eta\langle(w^* - \beta_k v_k - (1-\beta_k)\zeta_k, \nabla f(\zeta_k)\rangle + \gamma_k^2\eta^2\sigma^2 \\
&= \beta_k r_k^2 + (1-\beta_k)\|\zeta_k - w^*\|^2 + \gamma_k^2\eta^2\rho\|\nabla f(\zeta_k)\|^2 + 2\gamma_k\eta\langle\beta_k(\zeta_k - v_k) + w^* - \zeta_k, \nabla f(\zeta_k)\rangle + \gamma_k^2\eta^2\sigma^2 \\
&= \beta_k r_k^2 + (1-\beta_k)\|\zeta_k - w^*\|^2 + \gamma_k^2\eta^2\rho\|\nabla f(\zeta_k)\|^2 + 2\gamma_k\eta\left[\frac{\beta_k(1-\alpha_k)}{\alpha_k}\langle w_k - \zeta_k\rangle + w^* - \zeta_k, \nabla f(\zeta_k)\right] + \gamma_k^2\eta^2\sigma^2 \\
&\qquad \text{(From equation 4)} \\
&= \beta_k r_k^2 + (1-\beta_k)\|\zeta_k - w^*\|^2 + \gamma_k^2\eta^2\rho\|\nabla f(\zeta_k)\|^2 + 2\gamma_k\eta\left[\frac{\beta_k(1-\alpha_k)}{\alpha_k}\langle\nabla f(\zeta_k), (w_k - \zeta_k)\rangle + \langle\nabla f(\zeta_k), w^* - \zeta_k\rangle\right] + \gamma_k^2\eta^2\sigma^2 \\
&\leq \beta_k r_k^2 + (1-\beta_k)\|\zeta_k - w^*\|^2 + \gamma_k^2\eta^2\rho\|\nabla f(\zeta_k)\|^2 + 2\gamma_k\eta\left[\frac{\beta_k(1-\alpha_k)}{\alpha_k}(f(w_k) - f(\zeta_k)) + \langle\nabla f(\zeta_k), w^* - \zeta_k\rangle\right] + \gamma_k^2\eta^2\sigma^2 \\
&\qquad \text{(By convexity)}
\end{aligned}$$

By strong convexity,

$$\begin{aligned}
\mathbb{E}[r_{k+1}^2] &\leq \beta_k r_k^2 + (1-\beta_k)\|\zeta_k - w^*\|^2 + \gamma_k^2\eta^2\rho\|\nabla f(\zeta_k)\|^2 \\
&+ 2\gamma_k\eta\left[\frac{\beta_k(1-\alpha_k)}{\alpha_k}(f(w_k) - f(\zeta_k)) + f^* - f(\zeta_k) - \frac{\mu}{2}\|\zeta_k - w^*\|^2\right] + \gamma_k^2\eta^2\sigma^2
\end{aligned} \tag{16}$$

By Lipschitz continuity of the gradient,

$$\begin{aligned}
f(w_{k+1}) - f(\zeta_k) &\leq \langle\nabla f(\zeta_k), w_{k+1} - \zeta_k\rangle + \frac{L}{2}\|w_{k+1} - \zeta_k\|^2 \\
&\leq -\eta\langle\nabla f(\zeta_k), \nabla f(\zeta_k, z_k)\rangle + \frac{L\eta^2}{2}\|\nabla f(\zeta_k, z_k)\|^2
\end{aligned}$$

Taking expectation wrt z_k and using equations 9, 10

$$\mathbb{E}[f(w_{k+1}) - f(\zeta_k)] \leq -\eta\|\nabla f(\zeta_k)\|^2 + \frac{L\rho\eta^2}{2}\|\nabla f(\zeta_k)\|^2 + \frac{L\eta^2\sigma^2}{2}$$
$$\mathbb{E}[f(w_{k+1}) - f(\zeta_k)] \leq \left[-\eta + \frac{L\rho\eta^2}{2}\right]\|\nabla f(\zeta_k)\|^2 + \frac{L\eta^2\sigma^2}{2}$$

If $\eta \leq \frac{1}{\rho L}$,

$$\begin{aligned}
\mathbb{E}[f(w_{k+1}) - f(\zeta_k)] &\leq \left(\frac{-\eta}{2}\right)\|\nabla f(\zeta_k)\|^2 + \frac{L\eta^2\sigma^2}{2} \\
\implies \|\nabla f(\zeta_k)\|^2 &\leq \left(\frac{2}{\eta}\right)\mathbb{E}[f(\zeta_k) - f(w_{k+1})] + L\eta\sigma^2
\end{aligned} \tag{17}$$

From equations 16 and 17

$$\begin{aligned}
\mathbb{E}[r_{k+1}^2] &\leq \beta_k r_k^2 + (1-\beta_k)\|\zeta_k - w^*\|^2 + 2\gamma_k^2\eta\rho\mathbb{E}[f(\zeta_k) - f(w_{k+1})] \\
&+ 2\gamma_k\eta\left[\frac{\beta_k(1-\alpha_k)}{\alpha_k}(f(w_k) - f(\zeta_k)) + f^* - f(\zeta_k) - \frac{\mu}{2}\|\zeta_k - w^*\|^2\right] + \gamma_k^2\eta^2\sigma^2 + L\gamma_k^2\eta^2\rho\sigma^2 \\
&= \beta_k r_k^2 + (1-\beta_k)\|\zeta_k - w^*\|^2 + 2\gamma_k^2\eta\rho\mathbb{E}[f(\zeta_k) - f(w_{k+1})] \\
&+ 2\gamma_k\eta\left[\frac{\beta_k(1-\alpha_k)}{\alpha_k}(f(w_k) - f(\zeta_k)) + f^* - f(\zeta_k) - \frac{\mu}{2}\|\zeta_k - w^*\|^2\right] + 2\gamma_k^2\eta^2\sigma^2 \qquad \text{(Since } \eta \leq \tfrac{1}{\rho L}\text{)} \\
&= \beta_k r_k^2 + \|\zeta_k - w^*\|^2[(1-\beta_k) - \gamma_k\mu\eta] + f(\zeta_k)\left[2\gamma_k^2\eta\rho - 2\gamma_k\eta\cdot\frac{\beta_k(1-\alpha_k)}{\alpha_k} - 2\gamma_k\eta\right] \\
&- 2\gamma_k^2\eta\rho\mathbb{E}f(w_{k+1}) + 2\gamma_k\eta f^* + \left[2\gamma_k\eta\cdot\frac{\beta_k(1-\alpha_k)}{\alpha_k}\right]f(w_k) + 2\gamma_k^2\eta^2\sigma^2
\end{aligned}$$

Example - do not read!

Example - do not read!

Legitimate questions about performance analyses?

Lemma 3. *Assume that the function is L-smooth and μ strongly-convex and satisfies the strong-growth condition in Equation 10. Then, using the updates in Equation 3-5 and setting the parameters according to Equations 16-18, if $\eta \leq \frac{1}{\mu L}$, then the following relation holds:*

$$\mathbb{E}\|\tilde{x}_{k+1}\|^2 \leq \frac{\alpha_k^2}{\rho\eta}\left(f(x_0) - f^*\right) + \frac{\mathbb{E}}{2\rho\eta}\|x_0 - u^*\|^2 + \frac{\alpha^2\eta}{\rho}\sum_{t=0}^{k}\left[\gamma_t\|\tilde{h}_{t+1}^t\|\right]$$

Proof.

Let $r_{k+1} = \|u_{k+1} - u^*\|$, then using equation 5

$$r_{k+1}^2 = \|\beta_k v_k + (1-\beta_k)\zeta_k - u^* - \gamma_k\eta\nabla f(\zeta_k, z_k)\|^2$$
$$r_{k+1}^2 = \|\beta_k v_k + (1-\beta_k)\zeta_k - u^*\|^2 + \gamma_k^2\eta^2\|\nabla f(\zeta_k, z_k)\|^2 + 2\gamma_k\eta\langle u^* - \beta_k v_k - (1-\beta_k)\zeta_k, \nabla f(\zeta_k, z_k)\rangle$$

Taking expectation wrt to z_k,

$$\begin{aligned}
\mathbb{E}[r_{k+1}^2] &= \mathbb{E}[\|\beta_k v_k + (1-\beta_k)\zeta_k - u^*\|^2] + \gamma_k^2\eta^2\mathbb{E}\|\nabla f(\zeta_k, z_k)\|^2 + 2\gamma_k\eta\,\mathbb{E}[\langle u^* - \beta_k v_k - (1-\beta_k)\zeta_k, \nabla f(\zeta_k, z_k)\rangle] \\
&\leq \|\beta_k v_k + (1-\beta_k)\zeta_k - u^*\|^2 + \gamma_k^2\eta^2\rho\|\nabla f(\zeta_k)\|^2 + 2\gamma_k\eta\langle u^* - \beta_k v_k - (1-\beta_k)\zeta_k, \nabla f(\zeta_k)\rangle + \gamma_k^2\eta^2\sigma^2 \\
&= \|\beta_k(v_k - u^*) + (1-\beta_k)(\zeta_k - u^*)\|^2 + \gamma_k^2\eta^2\rho\|\nabla f(\zeta_k)\|^2 + 2\gamma_k\eta\langle(u^* - \beta_k v_k - (1-\beta_k)\zeta_k, \nabla f(\zeta_k)\rangle + \gamma_k^2\eta^2\sigma^2 \\
&\leq \beta_k\|v_k - u^*\|^2 + (1-\beta_k)\|\zeta_k - u^*\|^2 + \gamma_k^2\eta^2\rho\|\nabla f(\zeta_k)\|^2 + 2\gamma_k\eta\langle(u^* - \beta_k v_k - (1-\beta_k)\zeta_k, \nabla f(\zeta_k)\rangle + \gamma_k^2\eta^2\sigma^2 \\
&\qquad \text{(By convexity of } \|\cdot\|^2\text{)} \\
&= \beta_k r_k^2 + (1-\beta_k)\|\zeta_k - u^*\|^2 + \gamma_k^2\eta^2\rho\|\nabla f(\zeta_k)\|^2 + 2\gamma_k\eta\langle(u^* - \beta_k v_k - (1-\beta_k)\zeta_k, \nabla f(\zeta_k)\rangle + \gamma_k^2\eta^2\sigma^2 \\
&= \beta_k r_k^2 + (1-\beta_k)\|\zeta_k - u^*\|^2 + \gamma_k^2\eta^2\rho\|\nabla f(\zeta_k)\|^2 + 2\gamma_k\eta\langle\beta_k(\zeta_k - v_k) + u^* - \zeta_k, \nabla f(\zeta_k)\rangle + \gamma_k^2\eta^2\sigma^2 \\
&= \beta_k r_k^2 + (1-\beta_k)\|\zeta_k - u^*\|^2 + \gamma_k^2\eta^2\rho\|\nabla f(\zeta_k)\|^2 + 2\gamma_k\eta\left\langle\frac{\beta_k(1-\alpha_k)}{\alpha_k}(w_k - \zeta_k) + u^* - \zeta_k, \nabla f(\zeta_k)\right\rangle + \gamma_k^2\eta^2\sigma^2 \\
&\qquad \text{(From equation 4)} \\
&= \beta_k r_k^2 + (1-\beta_k)\|\zeta_k - u^*\|^2 + \gamma_k^2\eta^2\rho\|\nabla f(\zeta_k)\|^2 + 2\gamma_k\eta\left[\frac{\beta_k(1-\alpha_k)}{\alpha_k}\langle\nabla f(\zeta_k), (w_k - \zeta_k)\rangle + \langle\nabla f(\zeta_k), u^* - \zeta_k\rangle\right] + \gamma_k^2\eta^2\sigma^2 \\
&\leq \beta_k r_k^2 + (1-\beta_k)\|\zeta_k - u^*\|^2 + \gamma_k^2\eta^2\rho\|\nabla f(\zeta_k)\|^2 + 2\gamma_k\eta\left[\frac{\beta_k(1-\alpha_k)}{\alpha_k}(f(w_k) - f(\zeta_k)) + \langle\nabla f(\zeta_k), u^* - \zeta_k\rangle\right] + \gamma_k^2\eta^2\sigma^2 \\
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\end{aligned}$$

By strong convexity,

$$\begin{aligned}
\mathbb{E}[r_{k+1}^2] &\leq \beta_k r_k^2 + (1-\beta_k)\|\zeta_k - u^*\|^2 + \gamma_k^2\eta^2\rho\|\nabla f(\zeta_k)\|^2 \\
&\quad + 2\gamma_k\eta\left[\frac{\beta_k(1-\alpha_k)}{\alpha_k}(f(w_k) - f(\zeta_k)) + f^* - f(\zeta_k) - \frac{\mu}{2}\|\zeta_k - u^*\|^2\right] + \gamma_k^2\eta^2\sigma^2
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By Lipschitz continuity of the gradient,

$$\begin{aligned}
f(w_{k+1}) - f(\zeta_k) &\leq \langle\nabla f(\zeta_k), w_{k+1} - \zeta_k\rangle + \frac{L}{2}\|w_{k+1} - \zeta_k\|^2 \\
&\leq -\eta\langle\nabla f(\zeta_k), \nabla f(\zeta_k, z_k)\rangle + \frac{L\eta^2}{2}\|\nabla f(\zeta_k, z_k)\|^2
\end{aligned}$$

Taking expectation wrt z_k and using equations 9, 10

$$\mathbb{E}[f(w_{k+1}) - f(\zeta_k)] \leq -\eta\|\nabla f(\zeta_k)\|^2 + \frac{L\rho\eta^2}{2}\|\nabla f(\zeta_k)\|^2 + \frac{L\eta^2\sigma^2}{2}$$
$$\mathbb{E}[f(w_{k+1}) - f(\zeta_k)] \leq \left[-\eta + \frac{L\rho\eta^2}{2}\right]\|\nabla f(\zeta_k)\|^2 + \frac{L\eta^2\sigma^2}{2}$$

If $\eta \leq \frac{1}{\rho L}$,

$$\begin{aligned}
\mathbb{E}[f(w_{k+1}) - f(\zeta_k)] &\leq \left(\frac{-\eta}{2}\right)\|\nabla f(\zeta_k)\|^2 + \frac{L\eta^2\sigma^2}{2} \\
\implies \|\nabla f(\zeta_k)\|^2 &\leq \left(\frac{2}{\eta}\right)\mathbb{E}[f(\zeta_k) - f(w_{k+1})] + L\eta\sigma^2
\end{aligned} \tag{17}$$

From equations 16 and 17

$$\begin{aligned}
\mathbb{E}[r_{k+1}^2] &\leq \beta_k r_k^2 + (1-\beta_k)\|\zeta_k - u^*\|^2 + 2\gamma_k^2\eta\rho\mathbb{E}[f(\zeta_k) - f(w_{k+1})] \\
&\quad + 2\gamma_k\eta\left[\frac{\beta_k(1-\alpha_k)}{\alpha_k}(f(w_k) - f(\zeta_k)) + f^* - f(\zeta_k) - \frac{\mu}{2}\|\zeta_k - u^*\|^2\right] + \gamma_k^2\eta^2\sigma^2 + L\gamma_k^2\eta^2\rho\sigma^2 \\
&= \beta_k r_k^2 + (1-\beta_k)\|\zeta_k - u^*\|^2 + 2\gamma_k^2\eta\rho\mathbb{E}[f(\zeta_k) - f(w_{k+1})] \\
&\quad + 2\gamma_k\eta\left[\frac{\beta_k(1-\alpha_k)}{\alpha_k}(f(w_k) - f(\zeta_k)) + f^* - f(\zeta_k) - \frac{\mu}{2}\|\zeta_k - u^*\|^2\right] + 2\gamma_k^2\eta^2\sigma^2 \qquad \text{(Since } \eta \leq \tfrac{1}{\rho L}\text{)} \\
&= \beta_k r_k^2 + \|\zeta_k - u^*\|^2[(1-\beta_k) - \gamma_k\mu\eta] + f(\zeta_k)\left[2\gamma_k^2\eta\rho - 2\gamma_k\eta\cdot\frac{\beta_k(1-\alpha_k)}{\alpha_k} - 2\gamma_k\eta\right] \\
&\quad - 2\gamma_k^2\eta\mathbb{E}f(w_{k+1}) + 2\gamma_k\eta f^* + \left[2\gamma_k\eta\cdot\frac{\beta_k(1-\alpha_k)}{\alpha_k}\right]f(w_k) + 2\gamma_k^2\eta^2\sigma^2
\end{aligned}$$

Example - do not read!

Example - do not read!

- ⚠ Error-prone
- ⚠ Technical, lack global insights.
- ⚠ Few proof patterns.

- ? Easily fixable?
- ? Simple to adapt to variations of target inequality?
- ? Simple to adapt to algorithmic variations?

Organization and learning outcomes (1/3)

Observations:

- ◇ Complexity analyses / convergence proofs for first-order optimization methods often follow very similar (obscure?) patterns.
- ◇ ... these proofs are rarely intuitive—rely on combining (many) inequalities.
- ◇ Many variations around same themes.

Performance estimation framework:

- ◇ helps understand proof structures of worst-case analyses,
- ◇ provides principled way to explore complexity bounds/derive such analyses,
- ◇ mathematical proofs via numerical experiments and symbolic computations.

Organization and learning outcomes (2/3)

This mini-course:

- ◇ Interactive introduction to *performance estimation problems*.
- ◇ Learn how to use numerical PEPs to assess performance of simple first-order algorithms.
- ◇ Learn how to exploit PEP outputs to construct Lyapunov-based convergence proofs.
  - – Sub-goal: construct a proof for a simple gradient-like algorithm.
- ◇ Symbolic computations for search & verifying obtained guarantees.
  - – Sub-goal: construct a proof for variations around Nesterov's methods.
- ◇ Learn the basics of PEPs for numerical design of optimization algorithms.
  - – Sub-goal: design optimal methods in a few optimization setups.

Organization and learning outcomes (3/3)

0. Organization (now) ($\sim$ 10 mins)
1. Motivations, and a full (simple) example ($\sim$ 60 mins)
2. Missing details (and more!): ($\sim$ 60 mins)
   - first-order oracles, duality, interpolation,
   - semidefinite programming.
3. Guided hands-on notebook examples and Q&A ($\sim$ 60 mins)
4. Compact convergence & non-convergence certificates ($\sim$ 90 mins)
   - Using PEPs to search for cycles (no convergence).
   - Using PEPs to search for Lyapunov functions (convergence).
5. Guided hands-on notebook examples and Q&A ($\sim$ 90 mins)
6. Numerical design of optimized algorithms ($\sim$ 60 mins)
7. Guided hands-on notebook examples and Q&A ($\sim$ 60 mins)

missing 60mins — 3 blocks (base PEPs, Lyapunov-PEP, design)

Ressources

Ressources to be found at

`https://github.com/PerformanceEstimation/Tutorial-SMAI-MODE`

Please:

- do interrupt with questions!
- We will adapt the pace; program is probably optimistic.

Tools:

- Jupyter notebooks,
- Optimization tools/packages
  - CVXPY,
  - if possible: get academic MOSEK license (`https://www.mosek.com/products/academic-licenses/`)
- Symbolics:
  - SymPy
  - Mathematica is better (but expensive) — free (limited) version via cloud: `https://www.wolframcloud.com/`.